



Optimizing Manufacturing Sector through Business Process Reengineering and Lean Integration: A B2Lean Methodology Approach

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Abstract

This study aims to explore the integration of Business Process Reengineering (BPR) with Lean methodologies to optimize manufacturing processes. The proposed B2Lean methodology seeks to enhance efficiency, reduce waste, and improve overall performance in the manufacturing sector by merging the strategic approach of BPR with the continuous improvement focus of Lean practices. A case study methodology is employed to assess the implementation of the B2Lean approach in a manufacturing organization. The study involves a comprehensive analysis of existing processes, followed by the design and application of the B2Lean framework. Quantitative data on process performance before and after implementation are analyzed to evaluate the effectiveness of the integration. The findings reveal that the B2Lean methodology significantly improves operational efficiency, reduces cycle times, and enhances product quality. The synergy between BPR's radical process redesign and Lean's waste minimization leads to sustainable improvements in the manufacturing sector. Key success factors and challenges encountered during the implementation are also identified. The research provides valuable insights for manufacturing managers seeking to optimize their processes through a hybrid approach. The B2Lean methodology offers a structured framework for achieving substantial improvements in process performance, making it a practical tool for organizations aiming to stay competitive in a dynamic market environment. The integration of BPR and Lean has the potential to appeal to a broad audience, including those interested in operations management, supply chain management, and organizational change. This study contributes to the existing literature by presenting a novel approach that integrates BPR and Lean methodologies. The B2Lean framework offers a unique perspective on process optimization, combining the strengths of both approaches to deliver superior outcomes in the manufacturing sector.

Keywords Business Process Reengineering (BPR) · Lean integration · B2Lean methodology · IT- enabler · Continuous improvement · Manufacturing process optimization · Manufacturing sector

1 Introduction

In recent years, the manufacturing sector has encountered mounting pressure to boost operational efficiency and maintain competitiveness amidst a swiftly evolving global market. Traditional production and management techniques have proven inadequate to meet the demands of contemporary manufacturing, which necessitates a more streamlined, flexible, and effective approach [7, 25, 35]. This transformation is driven by the imperative to minimize waste, enhance process efficiency, and rapidly adapt to shifting customer demands—all essential for sustaining a competitive advantage [11, 14, 22]. To address these challenges, the integration of Business Process Reengineering (BPR) and Lean methodologies offers a promising solution. BPR focuses on the radical redesign of core business processes to achieve substantial gains in productivity, efficiency, and quality, while Lean principles concentrate on reducing waste and maximizing customer value [3, 11, 14, 15, 29]. Despite these methodologies' potential advantages, many manufacturing organizations face difficulties in their successful implementation due to the complexities involved in aligning them with existing processes [6, 10].

This study introduces the B2Lean methodology, an innovative approach that combines the strengths of BPR and Lean to optimize manufacturing processes. The B2Lean methodology addresses both the structural elements of manufacturing processes and incorporates advanced management techniques, measurement systems, and a value-driven perspective to enhance overall organizational efficiency [5, 33]. By considering both technical and human factors in process optimization, B2Lean aims to foster a more agile, responsive, and efficient manufacturing environment [8, 14]. In this context, effective time management and resource allocation are pivotal. As supported by various studies, time management is crucial for ensuring that manufacturing activities emphasize both effectiveness and efficiency [16]. The B2Lean methodology integrates these principles, offering a comprehensive framework for optimizing manufacturing processes and achieving sustainable performance improvements [9, 11].

This research will present the B2Lean process model and illustrate its application within a manufacturing setting, demonstrating its effectiveness in streamlining operations, reducing waste, and enhancing overall process efficiency. The study will assess the impact of the B2Lean methodology on key performance indicators such as production time, resource utilization, and product quality, providing valuable insights for manufacturing organizations aiming to improve their operational capabilities.

2 Literature Review

In the quest to stay competitive and meet the challenges of an ever-evolving market, the manufacturing sector has increasingly turned to process optimization. Business Process Reengineering (BPR) and Lean Manufacturing have emerged as pivotal methodologies in this effort, aiming to eliminate inefficiencies and enhance process effectiveness [5, 19]. BPR, initially introduced as a transformative approach to overhauling business processes, involves a fundamental reassessment of current practices to improve key performance indicators such as cost, quality, service, and speed [11, 15, 19, 23, 37]. From a socio-economic standpoint, BPR and Lean have played a significant role in reshaping the manufacturing industry by aligning processes with organizational objectives, minimizing waste, and improving service quality [3, 4, 29]. These methodologies have been effectively utilized to streamline production processes, shorten lead times, and elevate product quality in the manufacturing sector [3, 22, 27, 30]. This is particularly crucial as manufacturing firms face intensified competition and the need to adapt swiftly to changing market conditions [30, 35, 39].

Lean Manufacturing, originating from the Toyota Production System, focuses on the relentless elimination of waste (*muda*) across processes to enhance customer value [1]. The amalgamation of Lean principles with BPR has led to the creation of hybrid methodologies, such as the B2Lean approach. This integration combines the radical process reengineering capabilities of BPR with Lean's waste-reduction focus, fostering not only operational efficiency but also a culture of continuous improvement within organizations [11, 14, 33, 33].

To realize these transformations, manufacturing organizations must address both technical and organizational aspects. This involves reengineering production layouts, optimizing information systems, standardizing operational procedures, and improving worker collaboration [31, 34]. Such comprehensive adjustments are critical for fully leveraging the benefits of BPR and Lean integration [31, 34, 35]. Implementing methodologies like BPR and Lean also necessitates effective change management strategies to ensure successful adoption. Change management is essential for introducing new processes and structures while addressing employee resistance [6, 24, 26, 32, 36]. Each methodology poses distinct challenges based on the extent and nature of the changes, making it crucial for organizations to clearly define their objectives before initiating these transformations [6, 8, 17, 36, 38].

In summary, the effective integration of BPR and Lean methodologies, as demonstrated by the B2Lean approach, provides manufacturing organizations with a robust framework for process optimization, waste reduction, and improved operational efficiency. When properly applied and managed, these methodologies can significantly enhance the long-term competitiveness and sustainability of the manufacturing sector.

3 Methodology

This research introduces a novel hybrid methodology, B2Lean, aimed at optimizing the manufacturing sector by integrating Business Process Reengineering (BPR) and Lean Thinking principles. The BPR component leverages Information Technology (IT) as a transformative tool to achieve significant improvements in existing processes. In contrast, the Lean Thinking aspect focuses on minimizing waste in resources, whether in products, services, or operations. The B2Lean approach combines the strengths of these two methodologies to offer a comprehensive framework for process optimization. Table 1 presents a comparative analysis of the B2Lean methodology with traditional BPR and Lean Thinking methodologies, highlighting differences in phases, research implications, and tools employed.

Table 2 provides a detailed description of the six phases of the proposed B2Lean model, outlining the activities, techniques, and tools/Instruments used in each phase. Each phase comprises a series of sequential steps designed to optimize the manufacturing process.

3.1 Identifying and Analyzing the As-Is Process

In the first phase, the current application process, referred to as the As-Is process, was thoroughly examined using a combination of observations, preliminary reviews, discussions, brainstorming sessions, and document analysis. This approach helped gather the following information:

- a) **Components of the Manufacturing Process:** Detailed process flowcharts, processing times for each operation, the number of operators and machines involved, and the types of forms or documentation required.
- b) **Technology and Equipment Utilization:** Analysis of the current technology stack, machinery, and equipment used in the production line.
- c) **Operational Performance Data:** Monthly production reports, including output, downtime, defects, and throughput rates.
- d) **Internal Audit Reports:** Insights from internal audits identifying gaps and compliance issues within the manufacturing process.
- e) **Quality Assurance Guidelines:** Quality control measures, standard operating procedures, and adherence to the Client's Charter.
- f) **Key Performance Indicators (KPIs):** Metrics such as cycle time, productivity rates, defect rates, and on-time delivery rates.
- g) **Standard Operating Procedure (SOP) Manuals:** Documentation outlining the standard work procedures and safety protocols.

This information is critical in identifying the inherent inefficiencies and bottlenecks in the current manufacturing process. To further dissect these issues, the '5W and 1H' (What, Why, When, Who, Where, and How) technique is applied during brainstorming sessions. This approach provides a systematic examination of each

Table 1 Comparison of the proposed B2Lean methodology with BPR and lean thinking methodologies

Methodology	BPR (Business Process Reengineering)	Lean Thinking	B2Lean
Phases	1. Set vision and goals. 2. Identify business processes. 3. Investigate and evaluate the processes. 4. Implement Information Technology measures. 5. Prototype processes. 6. Implement changes.	1. Identify current values or processes. 2. Map the flow of values. 3. Create a streamlined flow of values. 4. Produce value flow based on customer demand. 5. Continuously pursue perfection.	1. Identify and analyze the As-Is process. 2. Identify and model the flow of values. 3. Identify & determine the processes to be redesigned. 4. Identify & assess existing information technology capabilities. 5. Create the To-Be process. 6. Evaluate the effectiveness of the To-be process in relation to the As-Is process.
Implications for research	Radically renew existing processes to optimize efficiency.	Enhance customer satisfaction by removing non-value-adding or wasteful processes or cause wastes.	Radically reform processes by eliminating wastes in the As-Is process that helps create the To-Be process.
Instrument	Information Technology	Value Stream Mapping and A3 Reporting	Information Technology and Value Stream Mapping.

Table 2 The six phases of the proposed B2Lean methodology

Phases	Activities	Techniques	Tools/Instruments
1. Identify and analyze the As-Is process	- Conduct discussions with officers to help them understand the current process (the As-Is process). - Review documents and current practices of existing processes. - Analyze the processes using the '5 W 1H' technique.	- Brainstorming. - Document review. - Flow charts. - Structured observation.	- Flow charts - Manual work procedure. - Client charter. - Monthly statistical reports. - Internal audit reports. - Key performance indicators - Interview questions.
	- List each sub-process involved in the application process from start to finish. - Model existing processes using diagrams to represent each process and their relationships (Value Stream Mapping). - Identify the needs of the As-Is process. - Propose to eliminate inefficient processes or those that do not add values.	- Focus group discussions (Brainstorming). - Structured observations. - Flow charts. - Modeling via activity diagrams. - Modeling via use case diagrams.	Highlight the As-Is process on paper using brainstorming technique.
3. Identify the processes to be redesigned.	- Identify the most important stages of the processes. - Recognize potential variables and prioritize the processes that need redesigning.	- Interviews. - Structured observations. - Flow charts. - Case diagrams. - Activity diagrams.	- Interview questions. - Checklists. - Mahjong papers. - Marker pens.
4. Identify the capabilities of existing information technology.	- Identify the IT facilities available in the unit. - Analyze the available IT resources that could support the new processes. - Determine the proficiency levels of staff in information systems and technology. - Provide additional briefing or offer training or courses for new processes.	- Structured observations. - Free observations. - Interviews.	- Interview questions. - Checklists.
5. Create the To-be process	- Develop the To-Be process by incorporating improvements made to the As-Is process. - Automate the new processes where applicable.	Activity diagrams.	Brainstorming.

Table 2 (continued)

Phases	Activities	Techniques	Tools/Instruments
6. Evaluate the effectiveness of the To-be Process in relation to the As-Is Process	Assess the effectiveness of the To-Be process, particularly in terms of time saved in each redesigned process.	- Interviews. - Structured interviews. - Activity diagrams.	- Interview questions. - Checklists. - Maiking papers. - Marker pens.

Table 3 Analysis of issues relating to the existing (As-Is) manufacturing processes using the ‘5W and 1H’ technique

5W and 1H	Issues
What	Production lead times are too long, resulting in delays in meeting customer demand.
Why	The production process is inefficient due to multiple handoffs and lack of streamlined workflows
Where	Different production stages are managed by separate teams in different departments, causing delays in communication and coordination.
When	Each time a batch of products moves from one stage of the production line to the next (e.g., assembly to quality check).
Who	Machine operators, assembly line workers, quality control inspectors, shift supervisors, and production managers.
How	• Operators receive raw materials and components from the warehouse. • Components are assembled according to specified procedures. • Assembled products are moved to quality control for inspection. • Quality control inspectors check for defects and ensure compliance with standards. • Defective products are sent back for rework; approved products move to the packaging stage. • Packaged products are then stored in the finished goods inventory. • Delays occur at each stage due to manual data entry, paperwork, and approval requirements.
Conclusion	The extended production lead time is due to an overly complex process structure and lack of integrated workflows, resulting in excessive time to complete production cycles.

process flow, revealing several critical weaknesses and areas for improvement, as summarized in Table 3.

3.2 Identifying and Modeling the Value Stream Mapping

In the Value Stream Mapping phase, key stakeholders involved in the manufacturing process, such as production managers, line supervisors, and machine operators, were invited for a focus group session. The goal of this session was to seek improvements by refining each of the steps involved in the production process. During this phase, each step was documented chronologically and mapped from start to finish. The As-Is Process was modeled using a swim lane activity diagram to visually represent the overall flow of the manufacturing process. This diagram effectively highlighted the beginning of the process, the activities and personnel involved, and the time taken to complete each stage.

After mapping the current process, the next step was to identify and eliminate waste. Discussions were held to determine whether to retain or eliminate specific processes, considering their impact on overall efficiency. Adherence to the eight waste categories specified in Lean methodology—defects, overproduction, waiting, non-utilized talent, transportation, inventory, motion, and extra-processing—was strictly enforced before making any changes to the process.

3.3 Identifying the Processes Requiring Redesign

After refining the current processes, the next step was to identify which processes needed improvement through redesign or modification. The following tasks were performed:

- a) Identify the most critical stages of the manufacturing processes.
- b) Recognize potential variables that contribute to inefficiencies or bottlenecks.
- c) Select the processes requiring redesign to enhance performance and reduce waste.
- d) Prioritize the selected processes for effective and timely implementation.

Identifying the processes within the manufacturing line that required redesign or modification was crucial to achieving the objectives of optimizing operations and increasing productivity. Other processes that could significantly impact overall efficiency and quality were also examined.

3.4 Identifying Available Information Technology (IT) Facilities

In this phase, the existing IT facilities within the manufacturing unit that could support the smooth implementation of redesigned processes were examined. The level of IT literacy among the personnel was also assessed to identify those who would require training to effectively perform the new processes. The existing IT infrastructure was reviewed with a focus on the following aspects:

- a) Adequate number of computers and workstations across the manufacturing unit.
- b) Provision of computers and digital tools to all relevant personnel.
- c) Stable network connectivity to ensure seamless data flow and communication.
- d) Compatibility of new processes with existing IT infrastructure and software systems.

3.5 Creating and Modeling the To-Be Process

In this phase, the To-Be Process was designed to address and eliminate the weaknesses identified in the As-Is process. The objectives of the redesigned process were as follows:

- 1. To expedite production cycle times.
- 2. To minimize downtime and delays.
- 3. To improve overall operational efficiency.
- 4. To enhance product quality and customer satisfaction.
- 5. To reduce operational costs.
- 6. To enable effective data sharing and communication across departments.
- 7. To minimize redundancies and rework.

8. To reduce waste in terms of materials, time, and energy.

The To-Be Process is aimed at eliminating the inefficiencies and waste identified in the As-Is process within the manufacturing unit. Through Business Process Reengineering (BPR), the necessary information systems and IT requirements were identified to enhance existing practices, effectively transforming the As-Is Process into the To-Be Process. In this transformation, IT serves as a crucial enabler for improving performance and driving lean integration within the manufacturing sector.

3.6 Evaluating the Effectiveness of the To-Be Process

In the final phase of the B2Lean methodology, the newly designed To-Be Process was evaluated to assess its effectiveness in optimizing the manufacturing operations. Each change or modification made to the As-Is process was evaluated based on its impact on reducing production cycle time, energy consumption, and operational costs within the manufacturing sector. A qualitative evaluation of the To-Be Process was conducted using a case study approach that involved interviews, checklists (comparing As-Is and To-Be Processes), flow charts, and several classified documents, such as standard operating procedure (SOP) manuals, Key Performance Indicator (KPI) reports, production logs, internal audit reports, and monthly performance reports.

Formal approval for conducting interviews was obtained after one of the senior production managers confirmed that the interview questions were suitable and relevant. The questions were designed to focus on the demographics of the interviewees and the weaknesses of the existing manufacturing processes. They also aimed to gather insights on potential improvements to current practices, which would be critical inputs for developing the To-Be Process. The time taken (measured in minutes) to complete each step of the As-Is process was analyzed through direct observations and interviews. Specifically, the evaluation measured the processing time from the initiation of a production batch to its completion and transfer to the next stage or relevant department.

4 Data Analysis and Results

This section presents the analysis and discussion on the eliminated processes, the comparison between the As-Is and To-Be processes, and the effectiveness of the new process.

4.1 As-Is Process Versus To-Be Process

The To-Be Process was developed based on a comprehensive review of the As-Is Process, highlighting the justifications for retaining or eliminating certain steps.

Table 4 Sub-processes of the As-Is process to be eliminated and their justifications

No.	Process	As-Is Process to be Eliminated	Justification
1.	Registration Process	a) Manual entry of production orders b) Manual rotation of job orders c) Manual issuance of production schedules	This manual process is time-consuming and is replaced with an automated production order system to streamline operations. This process is inefficient as job orders are manually logged and rotated, which can be automated to reduce time and labor costs. This process is redundant since production schedules can be generated automatically, reducing unnecessary paperwork.
4.	Production Intervention Process	a) Checking production schedules manually b) Manual recording of production data	Production schedules are checked manually in a logbook, which slows down decision-making and planning. An automated system would expedite this process. Repeated manual entry of production data (e.g., batch numbers, product codes, operator details) results in data redundancy and potential errors. This can be eliminated through integrated data systems.
6.	Quality Inspection Process	a) Manual entry of inspection data in Form A b) Manual entry of inspection data in Form B c) Manual entry of product information in quality reports	Quality control personnel manually re-enter inspection data into Form A, leading to duplication of effort. This process is replaced with a digital data capture system. Similarly, manual entry in Form B is also redundant and is replaced with automated data integration to reduce time and errors. Manually typing product information in quality reports is time-consuming and can be automated for efficiency.
9.	Approval Management Process	a) Manual data entry into spreadsheets for approval processes b) Manual follow-up on approval status	Considerable time is spent manually entering repetitive data into spreadsheets for approvals. This process is automated to speed up decision-making. Employees must manually contact other departments or use instant messaging apps to check the status of approvals, which is inefficient. An integrated system would allow real-time tracking of approval statuses.

The elimination of these steps was guided by the seven types of waste identified in Lean methodology: overproduction, waiting times, excessive inventory, unnecessary movements, unnecessary transportation, defects or errors, and inefficient processes. In this study, several processes were removed due to excessive waiting times, data entry errors, and redundant movements. Table 4 illustrates the processes that were eliminated from the As-Is Process and the justifications for their removal. The original As-Is Process comprised four sub-processes: the registration process, production intervention process, quality inspection process, and approval management process.

The transformation from the As-Is Process to the To-Be Process focuses on eliminating waste, reducing cycle times, improving data accuracy, and enhancing overall efficiency. Through automation and integration of IT systems, the manufacturing sector can achieve leaner operations and better resource management, aligned with the goals of the B2Lean methodology.

4.2 Modeling the As-Is Process with Use Case Diagrams

In the context of manufacturing process optimization, Use Case Diagrams are widely used in software engineering to model the functional requirements of newly optimized processes or systems. Due to their universal utility, these diagrams were employed to model the As-Is Process by analyzing all existing processes and scenarios in the manufacturing context. This includes the initial order registration process, production line intervention activities, quality inspection and reporting, and the approval management process. Figure 1 illustrates the current manufacturing workflow, including key personnel and their respective roles. The As-Is Use Case Diagram helped provide a comprehensive view of existing operations, allowing for a better understanding of areas requiring optimization.

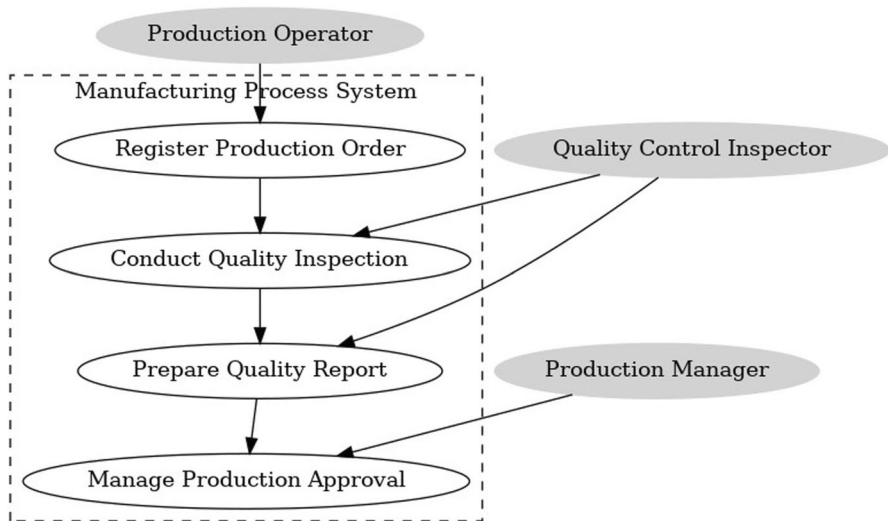


Fig. 1 As-Is use case diagram in a manufacturing sector context

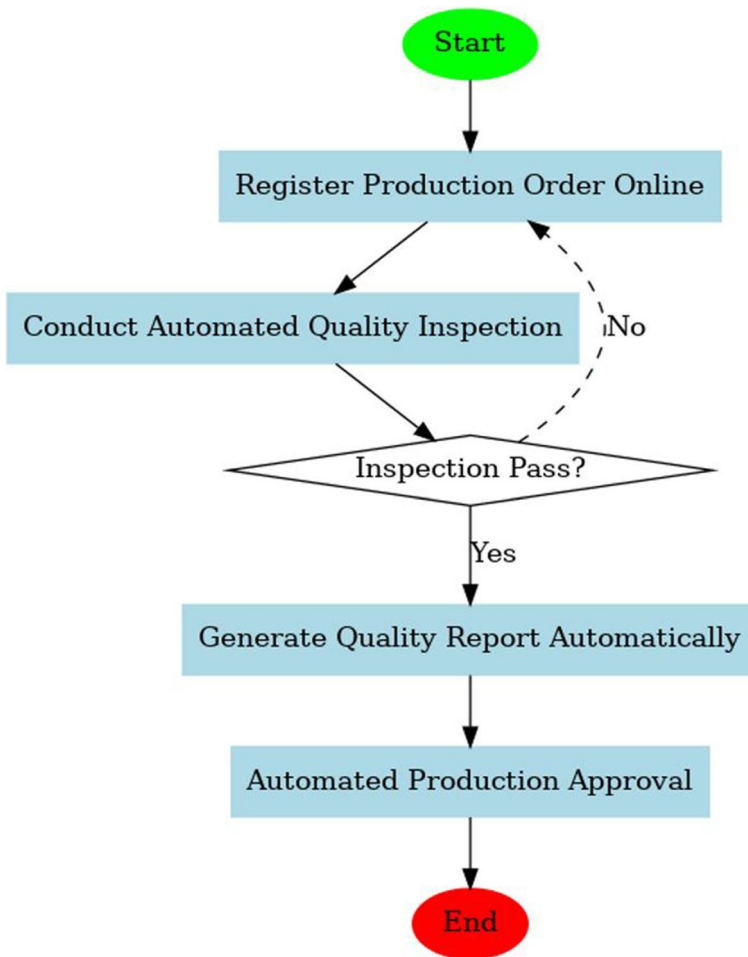


Fig. 2 To-Be process with activity diagrams in the manufacturing sector

4.3 Modeling the To-Be Process with Activity Diagrams

After identifying the processes needing improvement, the To-Be Process was modeled using Activity Diagrams to provide a visual comparison with the As-Is Process. Figure 2 shows the To-Be Process model, which has been refined by eliminating redundant, time-consuming steps and enhancing overall workflow efficiency. The new process model was mapped to show the flow of optimized processes, the involved personnel, and their specific roles. Figure 3 details the To-Be Process for production intervention, incorporating 10 streamlined processes and involving three key actors: the production operators, quality control inspectors, and production managers. This diagram represents the optimized processes, showing a streamlined workflow with automated steps for efficiency. A combined model (Fig. 2) is

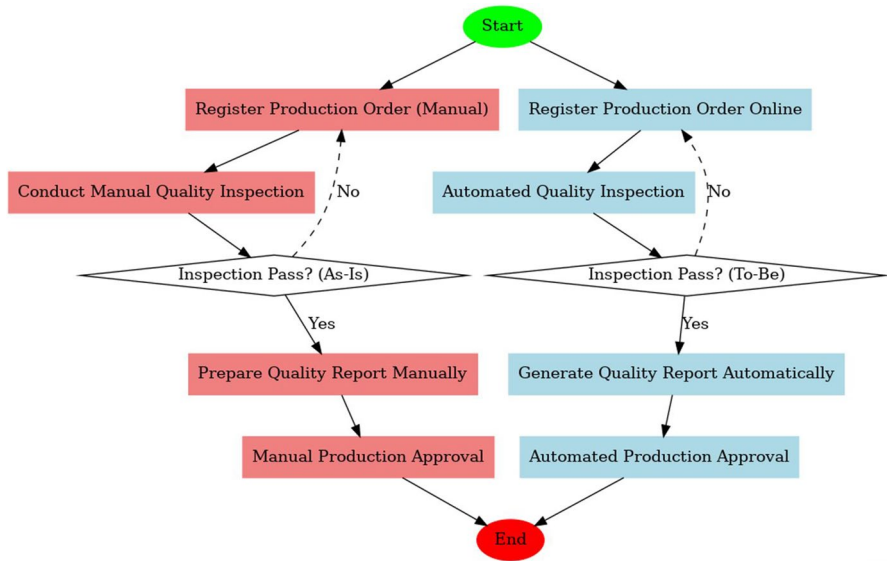


Fig. 3 Combined As-Is and To-Be activity diagram model

also given below to demonstrate the process improvement clearly. This model/diagram highlights the changes and optimizations from the current state (As-Is) to the improved state (To-Be).

4.4 Manufacturing Process Management System for Optimized Workflow

To eliminate redundant steps and enhance the efficiency of the As-Is process, the IT division within a manufacturing firm developed a comprehensive information system, named the Manufacturing Process Management System (MPMS). This system is designed to streamline tasks such as production order registration, scheduling, quality inspection, and reporting. MPMS supports the core functions of the To-Be process, which are aligned with the B2Lean methodology for optimizing manufacturing processes. The key functionalities of the MPMS are:

- **Online Production Order Registration:** Digitizes the registration process, reducing manual input errors and saving time.
- **Automated Production Scheduling and Intervention:** Efficiently schedules production activities and monitors process interventions, ensuring minimal downtime.
- **Automated Quality Report Generation:** Generates comprehensive quality reports automatically, reducing delays caused by manual reporting and ensuring data accuracy.

- **Integrated Approval Management System:** Manages approval workflows seamlessly, from production initiation to final quality clearance, ensuring all necessary validations are in place.

The system integrates multiple functions to ensure a seamless flow of information and processes across various departments, such as production, quality control, and inventory management. This integration allows quick access to real-time data for decision-making, enhancing overall process efficiency and reducing cycle times. The data and information stored in the MPMS can be accessed securely by authorized personnel anytime, anywhere, ensuring continuous monitoring and optimization of the manufacturing processes. Thus, Business Process Reengineering (BPR) and Lean integration, emphasizing the benefits of a robust information system like MPMS in achieving streamlined and efficient processes.

4.5 Evaluation of the To-Be Process

The primary objective of this study is to significantly reduce processing time, energy consumption, and operational costs in the manufacturing sector by implementing an optimized process. The newly designed To-Be process allows production teams and personnel to execute tasks more efficiently, ensuring leaner operations. An impact assessment was conducted to evaluate the effectiveness of the new process, focusing on the time required to complete all tasks involved in manufacturing workflows.

The deployment of the optimized process, supported by a new online management system, was executed in several phases, targeting specific bottlenecks identified in the As-Is process. These phases were developed based on removing redundant, labor-intensive tasks that previously slowed down operations. The new system incorporates streamlined scheduling, quality checks, and approval workflows to enhance overall efficiency.

Furthermore, the assessment also addressed challenges related to the existing IT infrastructure, such as outdated hardware and slow network connectivity, which negatively impacted the earlier processes. Upgrading these facilities was critical for the smooth functioning of the new To-Be process. By optimizing IT resources and integrating the process improvements, the study demonstrated a marked reduction in delays, resource wastage, and operational costs, aligning with the B2Lean methodology's goal of achieving sustainable efficiency in the manufacturing sector.

Therefore, the root causes of these inefficiencies were thoroughly investigated to ensure the successful implementation of the To-Be process, which comprises four newly improved sub-processes. As Murtuza (2015) emphasized, the application of Business Process Reengineering (BPR) can significantly enhance organizational workflows by reducing processing time and costs while improving product or service quality, ultimately leading to greater customer satisfaction.

In light of these potential benefits, the BPR approach was utilized to develop a more robust and effective process, designated as the To-Be process. This newly optimized process aims to enhance current practices within the manufacturing sector by streamlining operations, minimizing waste, and improving overall efficiency. The

Table 5 Processing Times of Sub-processes in the To-Be process and the As-Is process

No.	Name of sub-process	Processing time (in minutes)		Reduction in processing time (in minute)	Percentage of processing time used by TO-BE process	Percentage of processing time saved by TO-BE process
		AS-IS Process	TO-BE Process			
1.	Raw Material Handling	55	14	41	26%	74%
2.	Production Line Intervention	111	62	49	56%	44%
3.	Quality Control Report Preparation	248	108	140	43%	57%
4.	Report Submission	33	33	0	100%	0%
5.	Application Approval Process	80	37	43	46%	54%
6.	Total	527	254	273	48%	52%

To-Be process, grounded in the principles of Lean integration and B2Lean methodology, is poised to significantly elevate the quality of output and customer satisfaction by transforming the way manufacturing processes are managed and executed.

4.5.1 Differences in Processing Times between the As-Is and To-Be Processes

The comparative evaluation of the As-Is and To-Be processes was conducted to highlight the differences in processing times of their respective sub-processes. This evaluation was carried out through observations and interviews, as summarized in Table 5. For the production order registration process, the As-Is process and To-Be process have nine (9) and six (6) sub-processes, respectively, with processing times of 55 min and 14 min. This resulted in a significant reduction in processing time by 41 min, demonstrating the impact of Lean integration.

Similarly, a substantial reduction in processing time was achieved for the production intervention process, with the To-Be process reducing the time from 111 min to 62 min “a decrease of 49 minutes.” This improvement was made possible by reducing the number of sub-processes from 11 in the As-Is process to 10 in the To-Be process.

Furthermore, the preparation of manufacturing process reports showed remarkable efficiency improvements. The processing time was reduced from 248 min in the As-Is process to 108 min in the To-Be process, demonstrating that the To-Be process was more than twice as fast, despite both processes having the same number of sub-processes. Similarly, the process for approving production applications saw a significant reduction in processing time, from 80 min to 37 min, as the To-Be process streamlined operations to just six (6) sub-processes compared to 11 in the As-Is process.

However, no improvements were observed in the submission of production process reports due to constraints imposed by existing practices and policies that were beyond the control of the personnel involved.

The percentage of processing time used by the To-Be process is calculated as:

$$\text{Percentage of time used by To – Be process} = \frac{\text{To-Be process time}}{\text{As-Is process time}} \times 100$$

The percentage of processing time saved by the To-Be process is calculated as:

Percentage of time saved by To-Be process = 100 – Percentage of time used by To-Be process.

As depicted in Table 5, the To-Be process can significantly expedite the handling of raw materials by 74% compared to the As-Is process. It also substantially improves the production line intervention and quality control report preparation times by 44% and 57%, respectively. However, the report submission process could not be optimized further due to policy constraints requiring manual sign-offs, which cannot be replaced by automated methods. Finally, the To-Be process reduced the processing time of the application approval process by 54%. Overall, the newly proposed To-Be process reduced the total processing time by 52% compared to the existing process, marking a substantial improvement in manufacturing efficiency.

5 Discussion

In this study, a novel approach called B2Lean was introduced, integrating the principles of Business Process Reengineering (BPR) and Lean Thinking methodologies to optimize manufacturing operations. The aim was to streamline various manufacturing processes, reduce inefficiencies, and improve productivity and product quality. The development of this new method was guided by feedback obtained through group discussions with key stakeholders, such as production managers, line supervisors, and quality control personnel, who possess substantial experience in the current practices of manufacturing operations.

To gather relevant information on existing processes, the ‘5W 1H’ technique was employed, akin to the Zachman Framework, which can help detail a business process thoroughly [2, 21]. In addition, an analysis of existing documentation was conducted to uncover insights that could improve current manufacturing processes. Moreover, brainstorming sessions, field observations, and structured interviews were crucial in identifying gaps and inefficiencies in the As-Is process, laying the groundwork for the To-Be process improvements.

The Diamond Business System (DBS) model was adopted to conceptualize the B2Lean methodology. This model effectively visualizes the transformation that occurs when a company re-engineers its business processes [23]. As these scholars assert, one critical success factor in BPR is adhering to the principles of the DBS model, which clarifies the relationships between business processes, task and work structures, management and measurement systems, and organizational values and beliefs. In this context, business processes determine work structures, which in turn require a robust management and measurement system. Therefore, using this model within BPR significantly enhances existing processes in terms of cost, time, and quality [23]. Given these potential benefits, it is vital for manufacturing organizations struggling with high operating costs, low-quality products, or extended production times to consider BPR as a strategy for improving their current business practices [40].

The impact of the new To-Be process on accelerating manufacturing operations is potentially significant, as it can reduce overall processing times by more than half. In the manufacturing context, this reduction not only enhances production efficiency but also improves product quality and customer satisfaction by eliminating delays caused by redundant or unnecessary steps. This view aligns with [28], who emphasize that business processes comprise a set of necessary activities for transforming inputs into end-products or services that meet the needs of end-users. In this study’s context, these end-users are organizations seeking efficient and high-quality manufacturing processes.

However, introducing new practices is challenging because personnel may resist changes. Thus, manufacturing organizations aiming to implement organizational changes must employ sound change management strategies. According to Sturdy [36], change management is crucial for introducing newly designed processes and structures into work practices and effectively addressing employee resistance. This

involves thoroughly examining the purpose, level, and scope of change to help organizations focus on business processes needing improvement.

In this study, these processes were identified and outlined using standard Unified Modeling Language (UML) diagrams, such as use case diagrams and swimlane activity diagrams. The use case diagrams helped define the boundaries and scopes of the current tasks, while swimlane activity diagrams helped identify activities and tasks performed by specific personnel and their workflows throughout the production process. Moreover, the activity diagrams helped pinpoint and remove inefficient business processes characterized by overlapping or repetitive tasks. Such diagrams were effective in visualizing processes, enabling stakeholders to better understand and communicate how a business process is performing [18].

To ensure that new work structures are effectively implemented, a robust management system is needed. In the context of this study, the manufacturing process involves a series of tasks, from raw material handling to final product quality checks. Given the inefficiencies in current practices, an integrated system, referred to as the Manufacturing Process Management System (MPMS), was proposed as a tool to streamline production workflows by eliminating inefficient tasks. In today's digital age, such a system serves as a technological enabler, allowing manufacturing organizations to manage and control their activities more efficiently [20].

As demonstrated, stakeholders involved in implementing BPR can experience significant benefits from the new To-Be process, as it enables them to perform tasks more efficiently and with greater ease using MPMS. This system allows data and information to be more accessible through a centralized database, improving overall operational efficiency. The To-Be process, grounded in Lean principles, identifies and eliminates waste characterized by the seven waste types of Lean Thinking methodology, thus enhancing production efficiency. This improvement enables manufacturing personnel to experience more fulfilling and rewarding work environments.

Such benefits can be achieved through BPR, as this approach can enhance an organization's productivity by reducing processing times and costs and improving the quality of products or services, ultimately leading to greater customer satisfaction (Murtuza, 2015). Recently, Business Process Management (BPM) has emerged as a widely adopted approach for designing modern organizational and information systems. BPM's process-centered approach has shown to lead to substantial improvements in both system performance and compliance [18].

6 Conclusion

This study's findings highlight the substantial impact of implementing the B2Lean methodology on optimizing manufacturing processes. The proposed methodology, which integrates Business Process Reengineering (BPR) and Lean Thinking principles, aims to replace inefficient processes (the As-Is process) with streamlined and more effective processes (the To-Be process). The comparative modeling of the As-Is and To-Be processes demonstrated a significant reduction in overall processing time by more than 50%. Additionally, most manual and repetitive tasks were

automated using an integrated online system, referred to as the Manufacturing Process Management System (MPMS), to better meet the needs of stakeholders and practitioners. For instance, this system could enable real-time tracking of production processes, provide automatic alerts for equipment maintenance, and optimize inventory management through digital solutions.

In conclusion, it can be reasonably argued that adopting the new To-Be process based on the B2Lean methodology could significantly improve current manufacturing practices by enhancing efficiency, reducing waste, and improving product quality. These improvements not only boost operational performance but also enhance customer satisfaction and competitiveness in the market. Furthermore, the lessons learned from this study's application of the B2Lean methodology can serve as a guideline for other organizations seeking to refine their processes and adopt a more agile and efficient approach to business process management.

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Declarations

Competing Interests The authors declare no competing interests.

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Consent to Participate Informed consent was obtained from all participants involved in the study, ensuring their voluntary participation and understanding of the research's purpose. All participants provided consent for their anonymized data to be included in the publication. The authors have not intentionally engaged in or participate in any form that may be hateful to another person.

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